

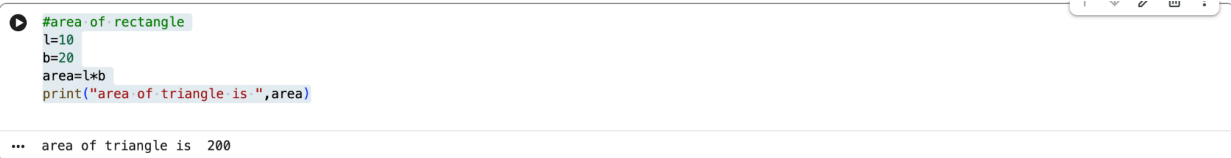
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1. Find area of rectangle.

Code:

```
#area of rectangle  
l=10  
b=20  
area=l*b  
print("area of triangle is ",area)
```

Output:



```
#area of rectangle  
l=10  
b=20  
area=l*b  
print("area of triangle is ",area)
```

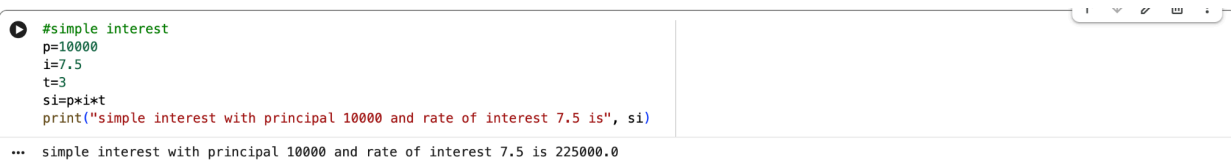
... area of triangle is 200

2. Find simple interest.

Code:

```
#simple interest  
p=10000  
i=7.5  
t=3  
si=p*i*t  
print("simple interest with principal 10000 and rate of interest 7.5 is", si)
```

Output:



```
#simple interest  
p=10000  
i=7.5  
t=3  
si=p*i*t  
print("simple interest with principal 10000 and rate of interest 7.5 is", si)
```

... simple interest with principal 10000 and rate of interest 7.5 is 22500.0

3. Convert temperature from Celsius to Fahrenheit.

Code:

```
# celcius to farhenite  
temp=32  
tempF=(temp*1.8)+32  
print("temperature in farhenite is",tempF)
```

Output:

```
# celcius to farhenite
temp=32
tempF=(temp*1.8)+32
print("temperature in farhenite is",tempF)

... temperature in farhenite is 89.6
```

4 .Calculate average of 3 numbers.

Code:

#average

num1=11

num2=12

num3=13

avg=(num1+num2+num3)/3

print("average of 11,12 and 13 is",avg)

Output:

```
#average
num1=11
num2=12
num3=13
avg=(num1+num2+num3)/3
print("average of 11,12 and 13 is",avg)

... average of 11,12 and 13 is 12.0
```

5. Find square and cube of a number.

Code:

#square and cube

num=4

sq=num**2

cube=num**3

print("square of 4 is",sq)

print("cube of 4 is",cube)

Output:

```
# square and cube
num=4
sq=num**2
cube=num**3
print("square of 4 is",sq)
print("cube of 4 is",cube)

... square of 4 is 16
   cube of 4 is 64
```

6. Swap two numbers without third variable.

Code:

#swap

a=3

b=4

a,b=b,a

print(f"a={a},b={b}")

Output:

```
#swap
a=3
b=4
a,b=b,a
print(f"a={a},b={b}")
... a=4, b=3
```

7 .Create a Student Report Program that take student details using input(), Store marks in variables, Calculate total and percentage , Use comments, Use proper indentation

Code:

```
def generate_report():
#collect student data
name=input("enter student name")
roll_no=input("enter roll number")
class_name=input("enter class name")
#Input Subject Marks
# input() captures everything as strings, so we convert them to float
print("\n please enter marks out of 100 for each subject \n")
math_marks=float(input("Mathematics:"))
english_marks=float(input("English:"))
science_marks=float(input("Science:"))
social_marks=float(input("Social Studies:"))
hindi_marks=float(input("Hindi:"))
# calculations
total=math_marks+english_marks+science_marks+social_marks+hindi_marks
max_marks=500
percentage=total/max_marks*100
print("\n" + "="*40)
print("      STUDENT REPORT CARD      ")
print("="*40)
print(f"Name:      {name}")
print(f"Roll No:   {roll_no}")
print(f"Class:     {class_name}")
print("-"*40)
print(f"Math Marks:  {math_marks:.1f} / 100")
print(f"Science Marks: {science_marks:.1f} / 100")
print(f"English Marks: {english_marks:.1f} / 100")
print("-"*40)
```

:.2f ensures the output formats cleanly to exactly two decimal places

```
print(f"TOTAL SCORE: {total:.2f} / {max_marks}")
print(f"PERCENTAGE: {percentage:.2f}%")
print("=*40)
generate_report()
```

Output:

```
... enter student nameJiya
enter roll number07801012025
enter class nameCSE 2

please enter marks out of 100 for each subject

Mathematics:95
English:96
Science:97
Social Studies:98
Hindi:94

=====
STUDENT REPORT CARD
=====
Name:      Jiya
Roll No:   07801012025
Class:     CSE 2
-----
Math Marks: 95.0 / 100
Science Marks: 97.0 / 100
English Marks: 96.0 / 100
-----
TOTAL SCORE: 480.00 / 500
PERCENTAGE: 96.00%
=====
```